

Development Automation **& Linux Lab**

Experiment 8

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EXPERIMENT-8

Working with Make and Makefile

1. Write a command to find out the log files and store it in another file using a single command.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques1$ ls
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques1$ find /var/log -name '*.log' -type f -exec cp {}
/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques1/ \;
find: '/var/log/private': Permission denied
find: '/var/log/unattended-upgrades': Permission denied
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques1$ ls
alternatives.log  dpkg.log          history.log        ubuntu-advantage.log
bootstrap.log     fontconfig.log    term.log          upgrade-policy-changed.log
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques1$
```

2. Write a program in C where there are two parts. The first part prints “Hello World” message. Store it in a file. The second part of the program directly display the error message of division by zero. Store the error message in another file.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ cat ques2.c
#include <stdio.h>
int main() {

    FILE *helloFile = fopen("hello.txt", "w");
    fprintf(helloFile, "Hello World\n");
    fclose(helloFile);

    int num = 1;
    int divisor = 0;
    FILE *errorFile = fopen("error.txt", "w");
    if (divisor == 0) {
        fprintf(errorFile, "Error: Division by zero\n");
    }
    fclose(errorFile);

    return 0;
}
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ gcc ques2.c -o ques2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ ./ques2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ ls
error.txt  hello.txt  ques2  ques2.c
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ cat error.txt
Error: Division by zero
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$ cat hello.txt
Hello World
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques2$
```

3. Write a program in C where there are two parts. The first part prints “Hello World” message. The second part of the program display the error message of division by zero. Store the two messages into a single file.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$ cat ques3.c
#include <stdio.h>
int main() {

    FILE *File = fopen("hello.txt", "a");
    fprintf(File, "Hello World\n");

    int num = 1;
    int divisor = 0;
    if (divisor == 0) {
        fprintf(File, "Error: Division by zero\n");
    }
    fclose(File);

    return 0;
}
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$ gcc ques3.c -o ques3
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$ ./ques3
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$ ls
hello.txt  ques3  ques3.c
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$ cat hello.txt
Hello World
Error: Division by zero
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques3$
```

4. Write a program in C where we can take input of two numbers and by using switch case we can do addition, subtraction, multiplication, and division. If the switch case is not '+', '-', '*' or '/' the some error is occurred. Store the error into a file. The whole switch case is done within while(1) loop. So, in future if some another error is occurred then add the errors into previously created error file.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques4$ cat ques4.c
#include <stdio.h>

int main() {
    double num1, num2, result;
    char op;

    while (1) {
        printf("Enter two numbers and an operator (+, -, *, /): ");
        scanf("%lf %lf %c", &num1, &num2, &op);

        switch (op) {
            case '+':
                result = num1 + num2;
                printf("%.2lf\n", result);
                break;
            case '-':
                result = num1 - num2;
                printf("%.2lf\n", result);
                break;
            case '*':
                result = num1 * num2;
                printf("%.2lf\n", result);
                break;
            case '/':
                if (num2 == 0) {
                    printf("Error: Division by zero\n");
                    FILE *errorFile = fopen("error.txt", "a");
                    fprintf(errorFile, "Error: Division by zero\n");
                    fclose(errorFile);
                } else {
                    result = num1 / num2;
                    printf("%.2lf\n", result);
                }
                break;
        }
    }
}
```

```

    default:
        printf("Error: Invalid operator\n");
        FILE *errorFile = fopen("error.txt", "a");
        fprintf(errorFile, "Error: Invalid operator\n");
        fclose(errorFile);
        break;
    }
}

return 0;
}

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques4$ gcc ques4.c -o ques4
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques4$ ./ques4
Enter two numbers and an operator (+, -, *, /): 1 4 *
4.00
Enter two numbers and an operator (+, -, *, /): 1 0 /
Error: Division by zero
Enter two numbers and an operator (+, -, *, /): 2 4 ^
Error: Invalid operator
Enter two numbers and an operator (+, -, *, /): 8 2 /
4.00
Enter two numbers and an operator (+, -, *, /): ^C
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques4$ cat error.txt
Error: Division by zero
Error: Invalid operator
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques4$

```

5. Try different Linux exit codes from the lab material.

- **exit 0:** exiting without any error.

```

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit1.sh
#!/bin/bash

echo "Hello, world!"
exit 0

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit1.sh
Hello, world!
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ echo $?
0
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$

```

- **exit 1:** catching all general errors.

```

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit1.sh
#!/bin/bash

echo "An error occurred."
exit 1

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit1.sh
An error occurred.
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ echo $?
1
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$

```

- **exit 126:** command invoked cannot execute.

```

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit126.sh
#!/bin/bash

# Try to execute a non-executable file
./non_executable_file.sh
exit 126

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit126.sh
./exit126.sh: line 4: ./non_executable_file.sh: No such file or directory
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$

```

- exit 127: command not found.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit127.sh
#!/bin/bash

# Try to execute a command that does not exist
unknown_command
exit 127

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit127.sh
./exit127.sh: line 4: unknown_command: command not found
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$
```

- exit 128: invalid argument to exit.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit128.sh
#!/bin/bash

# Attempt to run the script with incorrect arguments
if [ $# -lt 2 ]; then
    echo "Usage: $0 arg1 arg2"
    exit 128
fi

echo "Arguments: $1 and $2"

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit128.sh
Usage: ./exit128.sh arg1 arg2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$
```

- exit 130: script terminated by ctrl-c.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit130.sh
#!/bin/bash

# Trap the SIGINT signal (CTRL-C) and exit with exit code 130
trap 'echo "Interrupted by user"; exit 130' INT

# Wait for user input
read -p "Enter something: " input

# Print the user input
echo "You entered: $input"

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit130.sh
Enter something: 2
You entered: 2
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit130.sh
Enter something: ^CInterrupted by user
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$
```

- exit 255: exit status out of range.

```
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ cat exit255.sh
#!/bin/bash

# Check whether a required command is available
if ! command -v some_command &> /dev/null; then
    echo "Error: Required command 'some_command' not found"
    exit 255
fi

# Run the command
some_command --option argument

ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$ ./exit255.sh
Error: Required command 'some_command' not found
ayroid@Ayroid:/mnt/c/college/upes/sem4/linux/lab/scripts/exp8/ques5$
```